GNBF5020
Case Study Bank

English Edition · with Model Answers

Introduction to Molecular Biology and Genetics

20 exam-style scenario questions · Dr. Tony Wong · Lectures 1–7

Answer in English. Dr. Wong has stated clearly that every assignment and every exam in this course **must be answered in English** — last year he gave zero marks to students who answered in Chinese.
This is the **English edition**, so this is the copy you revise from: gene names, signalling pathways, drug names and detection methods must all be written in **English**.

Each case: vignette -> question -> model answer -> common trap

Companion document (Traditional Chinese): GNBF5020_案例研習題庫_繁體中文.pdf

Video guides

Sixteen terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index.html** plays all Sixteen on one page.

Video	Term	Chapter	Video file
V01	**Kaplan–Meier curve**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V01_Kaplan_Meier_curve.mp4
V02	**Contractile ring → binucleated cell**	Ch 8 · Cell biology	_video/assets/V02_Contractile_ring_binucleated_cell.mp4
V03	**NHEJ defect → SCID**	Ch 6 · DNA damage and repair	_video/assets/V03_NHEJ_defect_SCID.mp4
V04	**Mitochondrial pedigree**	Ch 8 · Cell biology	_video/assets/V04_Mitochondrial_pedigree.mp4
V05	**Meiosis I vs Meiosis II (M1 vs M2)**	Ch 8 · Cell biology	_video/assets/V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4
V06	**BER vs NER**	Ch 6 · DNA damage and repair	_video/assets/V06_BER_vs_NER.mp4
V07	**Hot tumour vs cold tumour**	Ch 9 · Immunology and cancer immunity	_video/assets/V07_Hot_tumour_vs_cold_tumour.mp4
V08	**Driver vs passenger**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V08_Driver_vs_passenger.mp4
V09	**IgG1 vs IgG4**	Ch 9 · Immunology and cancer immunity	_video/assets/V09_IgG1_vs_IgG4.mp4
V10	**Predictive vs prognostic biomarker**	Ch 11 · Assay techniques and omics	_video/assets/V10_Predictive_vs_prognostic_biomarker.mp4
V11	**cis vs trans**	Ch 7 · Control of gene expression	_video/assets/V11_cis_vs_trans.mp4
V12	**de novo vs maintenance methylation**	Ch 4 · Chromatin and epigenetics	_video/assets/V12_de_novo_vs_maintenance_methylation.mp4
V13	**DNA replication**	Ch 5 · DNA replication	_video/assets/V13_DNA_replication.mp4
V14	**Transcription**	Ch 7 · Control of gene expression	_video/assets/V14_Transcription.mp4
V15	**Translation**	Ch 2 · Proteins and enzymes	_video/assets/V15_Translation.mp4
V16	**DNA repair**	Ch 6 · DNA damage and repair	_video/assets/V16_DNA_repair.mp4

Open the video index page ->

Format mirrors the midterm short-question section: a clinical/genetic vignette + data + "what do you recommend, and why"

How to use these

- Read the vignette and the data **only** - write your answer before looking.
- Compare against the model answer, but the marks are for **reasoning**, not for matching wording.
- Check the **Common trap** line - that is where most marks are lost.
- Re-do any case you got wrong after 24 hours.

Every number in these cases comes from the lectures or the slides, so the reasoning transfers directly to the exam.

SECTION A - Biomarkers & drug selection (Lectures 1-2)

Case 1 - The resistant colon cancer patient

Vignette

A 58-year-old man presents with metastatic colorectal cancer. His tumour sequencing report shows:

- **KRAS G12D mutation - detected**
- **BRAF wild-type**
- **MSI-low** (no instability)

He asks whether he should be offered **cetuximab** (an anti-EGFR monoclonal antibody). He has read online that EGFR-targeted drugs "work well in colon cancer".

Question (a) Should he receive cetuximab? Justify using the signalling pathway.

Question (b) Name one molecular feature that *would* predict cetuximab benefit.

Question (c) His tumour is MSI-low, not MSI-high. Does immunotherapy apply?

MODEL ANSWER

(a) No. EGFR and RAS sit in the same linear pathway:

```
EGFR -> RAS -> RAF -> MEK -> ERK     (proliferation)
      -> PI3K -> AKT -> mTOR         (survival)
```

Cetuximab blocks **EGFR**, the receptor at the *top* of the cascade. Wild-type KRAS is a **transducer** - it only relays the signal while EGFR is firing, so blocking the receptor works. **Mutant KRAS is constitutively active**: it is stuck in the GTP-bound state and fires **regardless of whether EGFR is blocked**. The drug removes the upstream trigger but the downstream signal never stops, so the tumour keeps proliferating through RAF-MEK-ERK and PI3K-AKT-mTOR. This is a **predictive biomarker** failure - the mutation tells you the drug will not work, not that the patient is sicker.

(b) KRAS wild-type status. Optionally: BRAF wild-type, or (in other settings) EGFR amplification by FISH.

(c) No, not on this basis. Immunotherapy tracks **MSI/MMR**, not KRAS:

- **MSI-high** (defective MMR) -> frameshift neoantigen burden -> "hot tumour" -> responds to anti-PD-1/anti-PD-L1.
- **MSI-low** here means the mismatch-repair machinery is intact, so the neoantigen load is low -> "cold tumour" -> checkpoint blockade is unlikely to help.

Common trap: saying "KRAS mutation means cetuximab will help because it's a known mutation" - a mutation is only useful as a biomarker if it predicts response. Here it predicts *resistance*.

Case 2 - Reading a survival curve

Vignette

You find a Human Protein Atlas entry for **ERBB2 (HER2)** summarising three Kaplan-Meier plots, all at $p < 0.001$:

Cancer	Prognostic direction
Renal cancer	**Favourable** - high ERBB2 = better survival
Endometrial cancer	**Unfavourable** - high ERBB2 = worse survival
Pancreatic cancer	**Unfavourable** - high ERBB2 = worse survival

Question ERBB2 is a therapeutic target. Why would a pharma company not simply treat every ERBB2-high patient with Herceptin?

MODEL ANSWER

Because **prognostic** and **predictive** are different things:

- **Prognostic** = the marker tells you about outcome, whatever treatment you give.
- **Predictive** = the marker tells you whether a *specific treatment* will work for you.

A marker can be **prognostic but not predictive**. ERBB2 being favourable in renal cancer does not mean blocking it helps; it may simply be a correlate of a well-differentiated, less aggressive tumour. Only in settings where anti-ERBB2 therapy has a demonstrated survival benefit - breast and gastric - is ERBB2 used as a **predictive** biomarker. Elsewhere high ERBB2 is a description of prognosis and must not be read as a treatment recommendation.

Common trap: treating "high expression" as automatically equal to "treat it". Ask which evidence supports *the drug in this tumour type*.

Case 3 - A biomarker discovery you are handed

Vignette

A colleague proposes a **serum Vimentin** test for early hepatocellular carcinoma. Their stratified data:

Group	n	Serum Vimentin (ng/mL), mean +/- SD
Healthy controls	64	176.0 +/- 61.3
HCC, tumour <= 2 cm	44	271.5 +/- 165.9
HCC, tumour > 2 cm	44	231.3 +/- 132.3

Question (a) Which group best supports "early-stage" detection, and why does that matter?

Question (b) Give one reason this may still fail as a screening test.

Question (c) Which proteomics workflow produced the initial candidate list?

MODEL ANSWER

(a) The **<= 2 cm** group. It has the **largest separation from controls** (271.5 vs 176.0 ng/mL). A screening test is judged on how well it separates *early disease* from healthy people, not late disease - by the time a tumour is > 2 cm it is often already symptomatic, so a large signal there buys nothing. Interestingly the > 2 cm mean is **not higher** than the small-tumour mean, so the marker is not simply tracking tumour bulk.

(b) Any of: the **overlapping distributions** (SDs of 132-166 ng/mL against a 95 ng/mL mean gap mean individual patients will fall on the wrong side); the large **standard deviations**, which imply poor specificity; the need for a **cut-off ROC/AUC** before it can be used clinically; or the fact that this was measured in **already-diagnosed patients**, not a screening population - the false-positive rate in the general population is unproven.

(c) 2D-PAGE (2-DE) - first dimension **isoelectric focusing** (charge), second **SDS-PAGE** (molecular weight) - followed by excising the spot and identifying the protein by **LC-MS/MS**.

Common trap: quoting the mean difference alone. Always pair it with spread/overlap, and remember a diagnostic needs sensitivity, specificity and a defined threshold.

SECTION B - Cell biology & therapy platforms (Lecture 2)

Case 4 - The ambiguous flow plot

Vignette

A flow cytometry run on a patient's PBMCs produces a plot. Axes: **FSC vs SSC**, plus CD3 and CD4 fluorescence.

Question (a) What do FSC and SSC measure?

Question (b) A gate has been drawn around "CD3+CD4+ events". A colleague says "that's 40% helper T cells". Why is the gate essential, and what could go wrong without it?

Question (c) If the same stained sample must be recovered alive for downstream culture, which instrument is needed and why not a slide?

MODEL ANSWER

(a) FSC proportional to cell size; **SSC proportional to internal complexity / granularity** (roughness of the membrane and internal organelles).

(b) A leukocyte sample is a **mixture**. Raw acquisition includes lymphocytes, monocytes and granulocytes. The CD3/CD4 axes do not distinguish them cleanly, and monocytes and granulocytes can fall into overlapping regions and register partial fluorescence. **Gating on the lymphocyte region (FSC/SSC) first, then gating CD3+CD4+ within that population, is what makes the 40% interpretable.** Without gating, the number is contaminated by other cell types and can be badly wrong - which is exactly why the two **cyan** dyes are also specified.

(c) FACS (Fluorescence-Activated Cell Sorting) - the same optics plus electrostatic or mechanical droplet deflection to physically divert live cells into collection tubes. A slide (or a plain analyser) destroys the sample: it stains, fixes and mounts it. If you only want to *analyse*, an analyser is enough; if you need viable cells back for culture, sequencing or therapy, you need a sorter under sterile conditions.

Common trap: quoting a percentage without stating the gating hierarchy. "40% of PBMCs" and "40% of lymphocytes that are CD3+CD4+" are different numbers.

Case 5 - Manufacturing a CAR-T that keeps failing

Vignette

A team engineers CAR-T cells against a HER2-positive gastric tumour. After infusion the cells expand poorly and the tumour does not respond. Investigation shows the patients had received heavy prior chemotherapy.

Question (a) Name the three CAR-T manufacturing routes.

Question (b) Why does prior chemotherapy hurt autologous CAR-T performance?

Question (c) The team proposes an antigen-independent "safety switch" built from a **Notch** receptor. Explain how this changes the safety profile.

MODEL ANSWER

(a) Autologous (patient's own T cells), **allogeneic** (off-the-shelf, from healthy donors or iPSC-derived lines), **in vivo** (mRNA in lipid nanoparticles delivered to the patient's own T cells in the body, so they are engineered in place).

(b) Chemotherapy depletes and **dysfunctions the patient's T-cell compartment** - *T-cell fitness* (proliferative capacity, viability, memory phenotype) is poor. Autologous CAR-T is literally made from those damaged cells, so you are dose-limiting a product built from the worst available substrate. Allogeneic cells sidestep this because they come from a healthy, undamaged donor.

(c) A native **Notch** receptor is cleaved by ligand engagement, releasing **NICD**, which enters the nucleus as a transcription factor. Rewriting this into **synNotch** puts an inducible promoter under the control of that transcription event. This builds an **AND gate**: the CAR gene is expressed **only when antigen A (tumour cell) and antigen B (a contextual marker) are both present**.

Why it matters: a conventional CAR fires when it touches *any* cell carrying the target antigen, including healthy tissue expressing it weakly - the source of on-target/off-tumour toxicity. synNotch requires the tumour's *context* to be present as well, so normal tissue expressing antigen A alone cannot trigger the CAR. Targeting specificity rises sharply.

Common trap: describing synNotch as simply "a better CAR". The key is that it changes the **logic** from OR to AND.

Case 6 - Choosing a companion diagnostic

Vignette

An ADC trial is enrolling gastric cancer patients. The protocol requires HER2 status by **FISH**, not IHC. Two patients both show HER2-positive tumour by immunohistochemistry.

Question (a) What is FISH measuring that IHC is not, and why does that matter for anti-HER2 therapy?

Question (b) One patient is HER2-positive by IHC but FISH-negative. What do you conclude about her expected benefit?

MODEL ANSWER

(a) FISH uses a **fluorescent DNA probe** complementary to the HER2 locus, and counts **gene copies per nucleus**. So it detects **amplification** - a genomic change. IHC uses an **antibody** to detect HER2 **protein**. This matters because anti-HER2 therapy in gastric cancer is licensed against **amplified** HER2; protein can be expressed at high level in a cell that has not amplified the gene, and that protein-high / copy-low situation behaves differently.

(b) She should **not** be expected to benefit. Her tumour is protein-positive without amplification. This is exactly why the protocol specifies FISH: the trial's validated biomarker is gene amplification, and IHC alone would enrol patients outside the evidence base. It also illustrates why a cheap, available assay is not automatically the right one.

Common trap: saying "HER2 is positive, therefore treat". Positivity defined by the *wrong modality* is not the same indication.

SECTION C - Epigenetics & chromosomal disease (Lecture 3)

Case 7 - The glioblastoma treatment decision

Vignette

Two glioblastoma patients, same age, same WHO grade, same surgical extent.

- **Patient A:** MGMT promoter **hypermethylated**
- **Patient B:** MGMT promoter **unmethylated**

Both are about to receive temozolomide plus radiotherapy.

Question (a) Whose tumour will be harmed more by temozolomide, and why?

Question (b) Which patient should have better survival, and what is this biomarker called?

Question (c) Name the two classes of methyltransferase that maintain the mark.

MODEL ANSWER

(a) Patient A - and that is *good* news, not bad. Temozolomide methylates guanine at O6. The repair enzyme **MGMT** removes that methyl group. If the **MGMT promoter is methylated, MGMT is silenced**, so the O6-methylguanine lesion persists and is converted into a lethal cross-link/kill event. **Unmethylated MGMT (Patient B)** efficiently repairs the damage, so temozolomide does much less.

(b) Patient A has the better prognosis - MGMT promoter methylation is a **predictive biomarker** for temozolomide response (and it happens to be prognostic too). The clinical shorthand is "MGMT methylation predicts benefit from alkylating chemotherapy".

(c)

- **De novo methyltransferases - DNMT3A / DNMT3B:** establish brand-new marks on previously unmethylated CpG.
- **Maintenance methyltransferase - DNMT1:** reads the hemi-methylated parent strand after replication and copies the pattern onto the new strand.

Common trap: assuming "silenced gene = worse outcome". Here silencing a *repair* gene is exactly what makes the chemotherapy work.

Case 8 - A family pedigree and aneuploidy

Vignette

A couple have a child with **Down syndrome (Trisomy 21)**. Karyotyping shows 47 chromosomes. A second couple has a child with **Turner syndrome (45,X)**.

Question (a) At which meiotic stage did each error most likely occur?

Question (b) For Down syndrome, name the two possible origins.

Question (c) Why are the live-born outcomes so different?

MODEL ANSWER

(a)/(b) Both arise from **non-disjunction** - failure of chromosomes to separate - but at different divisions:

- **Trisomy 21** -> gamete has **24** chromosomes, so zygote is 47.
- **Turner 45,X** -> gamete has **no sex chromosome**, so zygote is 45,X.

- **Meiosis I error:** homologous chromosomes fail to separate -> produces **two disomic gametes + two nullisomic** gametes -> fertilisation by a normal gamete gives **trisomy** (n+1) or **monosomy** (n-1).
- **Meiosis II error:** sister chromatids fail to separate -> **one disomic + one nullisomic + two normal** -> also trisomy/monosomy in a different proportion.

Down syndrome typically arises from a **maternal meiosis I** error. Either parent can in principle be the source.

(c) The asymmetry is about **which chromosomes survive**. In trisomy 21 the embryo keeps 47 chromosomes, a small surplus of a small chromosome, and survives. In monosomy X the embryo is missing an entire sex chromosome - every X-linked gene is present in only one copy - and only a fraction of such conceptions survive; most miscarry. The X chromosome is unusually gene-dense and dosage-sensitive.

Common trap: saying "Turner is trisomy-like". It is a **monosomy**, and the live-birth rate differs because a whole chromosome is missing.

Case 9 - An inversion that changes expression without changing dosage

Vignette

A patient carries a large **inversion** on chromosome 7 near the *SHH* locus. Sequencing confirms the gene is intact - **copy number is unchanged**.

Question The patient has an abnormal phenotype. If no gene was gained, lost or mutated, how can expression change? What class of variant is this?

MODEL ANSWER

An inversion does not change gene **dosage** - nothing is added or removed. But it **changes genomic context**. Transcriptional regulation depends on physical proximity: an **enhancer** works by contacting a promoter, and this requires the two to be near each other in 3D chromatin. A large inversion can:

- **Move a gene away from its enhancer**, weakening or abolishing expression; or
- **Move a gene into contact with a different, inappropriate enhancer**, causing ectopic expression.

This is a classic example of a **structural variant that is pathogenic without being coding**, and it explains why copy-number analysis (CNV calling, FISH, karyotyping) alone would have reported this patient as normal.

Common trap: concluding "no mutation, therefore no effect". The variant class to name here is a **structural variant (inversion) affecting regulatory context**.

SECTION D - Replication, repair and targeted therapy (Lecture 4)

Case 10 - A tumour that resists the obvious drug

Vignette

A patient with high-grade serous ovarian cancer has a **BRCA1** germline mutation. Her tumour responds to a PARP inhibitor for 18 months, then progresses.

Question (a) What does a PARP inhibitor do in a BRCA1-deficient cell? Explain using synthetic lethality.

Question (b) Why does resistance develop?

Question (c) What is the therapeutic-window advantage here?

MODEL ANSWER

(a) Synthetic lethality: loss of *gene A* alone is survivable; loss of *gene B* alone is survivable; losing **both** kills the cell.

- The tumour is already **A-deficient** (BRCA1 loss -> defective homologous recombination).
- **HR is high-fidelity but has a backup.** Single-strand breaks are normally repaired by PARP-dependent pathways. PARP inhibition traps PARP on DNA, preventing repair of these single-strand breaks. They convert into **double-strand breaks**.
- Normal cells still have working HR and clear the damage. The BRCA1-deficient cells **cannot** - they have no HR to fall back on -> cell death.

So the drug selectively kills the cells that are *already* broken in the exact pathway the drug disables.

(b) Secondary reversion / restoration of HR. Resistance typically arises through:

- **Reversion mutations** that restore BRCA1 open reading frame/function;
- **Loss of 53BP1, REV7 or Shieldin** - these normally block end resection, and removing them restores end resection and therefore HR in the BRCA1-deficient cell.

Either way the cell **regains homologous recombination** and is no longer selectively killed.

(c) Normal cells have intact HR, so they tolerate the drug at doses that kill the tumour -> a **wide therapeutic window** between the effective dose and the toxic dose. This is why PARP inhibitors are far better tolerated than, say, classic cytotoxic DNA cross-linkers.

Common trap: saying "PARP inhibitors damage DNA so they kill tumour cells". The selectivity comes from **synthetic lethality**, not from DNA damage per se.

Case 11 - Interpreting an MMR result

Vignette

A colorectal cancer specimen is tested:

- **MLH1, MSH2, MSH6, PMS2** - all retained by IHC
- Microsatellite panel: **stable**

Question (a) Is this tumour MSI-high? What does that imply about the immune environment?

Question (b) Which targeted therapy class becomes relevant, and by what mechanism?

Question (c) Which circulating marker would a clinician track alongside imaging, and why is a surrogate useful here?

MODEL ANSWER

(a) No - this is MSS (microsatellite stable). IHC showing all four MMR proteins retained, plus a stable microsatellite panel, means the repair machinery is intact. Consequently there are few slippage-induced **frameshifts**, so the **neoantigen load is low**: a "**cold tumour**" with little T-cell infiltration and poor response to immunotherapy.

(b) Contrast this with **MSI-high** (MMR loss -> unrepaired indels at microsatellites -> frameshift mutations -> abundant tumour-specific neoantigens -> "hot tumour" -> responds to **immune checkpoint blockade**, e.g. anti-PD-1 or anti-PD-L1). Here that pathway is closed. Note also that **PD-L1** can still suppress any T cells that do infiltrate, which is why PD-1/PD-L1 blockade is tried in the MSI-H setting and generally not here.

(c) CEA (carcinoembryonic antigen). It is a fetal glycoprotein re-expressed by many colorectal tumours and is measurable in blood. It serves as a **surrogate marker**: repeated CEA gives an indication of tumour burden without the cost, radiation exposure, logistical difficulty and patient stress of serial CT/MRI. Because CEA can also be raised in benign conditions, it is tracked as a **trend**, and changes are interpreted alongside imaging rather than in isolation.

Common trap: equating "MMR retained" with "no immunotherapy ever". The correct reasoning here is that the neoantigen-generating pathway is intact, so the expected benefit is low.

Case 12 - Choosing an editing strategy

Vignette

A lab must achieve one of three goals in a human cell line:

- **Temporarily** reduce *GATA1* expression for 48 hours, then restore it.
- **Permanently** inactivate *GATA1* for the life of the cell.
- **Insert** a fluorescent tag at the exact start codon of *GATA1*, leaving the protein sequence unchanged.

Question For each goal, name the most appropriate tool and state which DSB-repair pathway does the work.

MODEL ANSWER

Goal	Tool	Repair pathway	Why
1. Temporary knockdown	**siRNA / shRNA**	none - RISC degrades the mRNA	You want to *reduce transcript*, not change the genome. Effect fades as the RNA is degraded and the construct dilutes.
2. Permanent inactivation	**CRISPR/Cas9**	**NHEJ**	Cas9 cuts both alleles. Error-prone NHEJ repairs the DSB with small indels; if the indel falls in the coding region it frameshifts the gene out. Because the genome is changed, the effect is inherited by every daughter cell.
3. Precise tag insertion	**CRISPR/Cas9 + donor template**	**HR** (homology-directed repair)	Supply a donor plasmid carrying the tag flanked by homology arms. The cell copies the donor into the cut site, placing the tag exactly at the start codon without altering the protein sequence.

Critical caveat: HR is **restricted to S/G2**, because a sister chromatid must exist to serve as the template. A confluent, serum-starved, G0-arrested culture - which is exactly how many cell lines are kept - will do NHEJ and give you indels instead of a clean knock-in. For goal 3 the cells must be **actively cycling and in S/G2**.

Common trap: using siRNA for a permanent knockout, or expecting a knock-in from a quiescent culture.

Case 13 - Why the topoisomerase inhibitor is in the antibody

Vignette

A new ADC, **trastuzumab-deruxtecan (T-DXd)**, is approved for HER2-positive breast cancer. It delivers **deruxtecan (DXd)**, a **topoisomerase I inhibitor**, linked to an anti-HER2 antibody.

Question (a) What does topoisomerase I normally do, and why is inhibition lethal?

Question (b) Why attach such a payload to an antibody at all, instead of giving free DXd?

Question (c) Several tumours relapse after responding well. Name one mechanism.

MODEL ANSWER

(a) During replication, helicase unwinds the helix, and **topoisomerase I** relieves the torsional strain that builds ahead of the fork by temporarily nicking one strand and re-ligating it. **Camptothecins and their derivatives trap the enzyme in the covalent cleavage complex**: the nicked DNA cannot be re-ligated, so each replication attempt converts a transient single-strand nick into an irreversible **double-strand break**, which the cell cannot repair -> death. It is a replication-dependent poison: it does little to non-dividing cells, which is part of why the therapeutic window exists.

(b) The antibody is a **targeting device**: it delivers a very potent payload selectively to HER2-expressing cells, sparing normal tissue. Free DXd would be distributed everywhere and would be limited by systemic toxicity. The strategy converts a drug that is far too toxic to use alone into one that can be dosed.

(c) Any of:

- **HER2 amplification ceases / antigen is lost** - no target, so the payload never arrives.
- **Topoisomerase I mutations or reduced expression** - the target of the payload is gone.
- **HER2/HER3 or HER2/HER4 heterodimer switching** - anti-HER2 binding that alters the dimer partner.
- **Efflux pumps (e.g. ABCB1)** exporting the payload.
- **Single-strand-break repair upregulation**.

Common trap: confusing topoisomerase I with topoisomerase II - the POIs used in ADCs (DXd, SN-38) target **I**; etoposide/anthracyclines target **II**.

SECTION E - Regulation of gene expression (Lecture 5)

Case 14 - A compound that switches itself off in the right cells

Vignette

A team is engineering CAR-T cells so they activate **only inside the tumour core**. The tumour core is hypoxic; normal tissue is not. They place the CAR transgene under the control of a **Hypoxia Response Element (HRE)** promoter.

Question (a) Explain why an HRE-controlled CAR stays quiet in normal tissue.

Question (b) The construct fails and CAR-T kill normal cells. What single upstream step did they most likely get wrong?

Question (c) How could they add a second, independent safety layer, and what logic would it give?

MODEL ANSWER

(a) The mechanism runs through **HIF-1alpha**.

- **Normoxia:** HIF-1alpha is continuously produced but immediately **degraded** (via prolyl hydroxylation -> VHL recognition -> ubiquitin-proteasome). It never accumulates, so it cannot activate the HRE. The CAR gene is **off**.
- **Hypoxia:** the hydroxylases cannot work, so HIF-1alpha **stabilises**, dimerises with HIF-1beta, enters the nucleus and binds the **HRE**.
- Therefore the CAR protein is produced **only where oxygen is low** - i.e. in the tumour core. The hypoxia response element is being used as a **molecular oxygen sensor**.

(b) The most likely failure is a **constitutive HIF-1alpha** problem - e.g. HIF-1alpha not properly degradable, or an HRE-driven promoter with a **leaky/basal activity**, or a **VHL-pathway defect** in the T cells. If HIF-1alpha is not degraded in normoxia, the gate is open everywhere and the CAR is expressed in all cells. The CAR then fires on any normal cell expressing the antigen - classic **on-target, off-tumour toxicity**.

(c) Add a **synNotch AND gate**: put the HRE-driven HIF response *upstream of a synNotch element* rather than driving the CAR directly.

- synNotch requires antigen **A** (tumour cell) and antigen **B** (a contextual marker) before the CAR gene is transcribed.
- Combined with the hypoxia input, the CAR is expressed only when **(tumour cell present) AND (tumour context present) AND (hypoxic)** - three inputs instead of one. Any single failure is survivable.

Common trap: using one environmental cue alone as the sole safety gate. Single-gate designs are fragile; the lecturer's point is that you stack independent orthogonal controls.

Case 15 - The isoform that predicts drug resistance

Vignette

In liver cancer, sequencing finds a **single-nucleotide substitution** that creates a **novel splice site**. The resulting protein lacks part of a domain and is secreted rather than membrane-bound. Patients whose tumours express it show significantly **worse overall survival**; mice overexpressing it show larger, more invasive tumours.

Question (a) What mechanism produced this change - one gene, but how many proteins?

Question (b) Name two clinical consequences of this event.

Question (c) If you designed an ASO to block it, what would it silence, and how is that different from blocking with an anti-sense DNA gap-mer?

MODEL ANSWER

(a) Alternative splicing. The spliceosome chooses splice sites from the pre-mRNA. A single base substitution can:

- create a **new splice acceptor/donor site**, or
- abolish a normal site,

so the exon is included, skipped or partially included in a different combination. One gene therefore yields a different **isoform** - and here a truncated, **soluble** version that no longer has the transmembrane domain, so it leaks out of the cell.

(b) Two valid consequences:

- **A secreted biomarker.** Because the isoform lacks the membrane anchor, it is measurable in **serum** - turning a tissue event into a non-invasive liquid-biopsy test for prognosis.
- **A neoantigen / immune escape route.** The novel isoform-derived peptides are tumour-specific. They can generate neoantigens that alter immunogenicity - helping convert a "cold tumour" to a "hot" one or vice versa - and, in the drug context, removing or truncating the **drug-binding domain** produces **treatment resistance**.

(c) An **ASO (antisense oligonucleotide)** is a short single-stranded DNA/RNA analogue that binds the complementary RNA and recruits RNase H, degrading the target mRNA. It therefore silences **both** the normal and the aberrant isoform - it acts pre-mRNA/mRNA.

An **anti-sense DNA gap-mer** binds a specific region (usually across the splice junction or an exon-intron boundary) and **blocks the spliceosome**, so only the intended isoform is suppressed. It is more selective - it spares the normal protein, which matters when you do not want to lose the wild-type function.

Common trap: describing alternative splicing as a mutation that damages the gene. The gene sequence changed by one base, but the mechanism that produces many proteins is **splicing**, and the therapeutic handle is the **splice site**, not the coding change.

Case 16 - Reading the lac operon logic correctly

Vignette

An expression construct is engineered using the bacterial lac system for *E. coli* expression of a human protein. The operator is disrupted in one clone and intact in another. IPTG is used to induce.

Question (a) What is IPTG, and why is it used instead of lactose in an industrial setting?

Question (b) In the operator-disrupted clone, transcription is constitutive. Give one production advantage and one production problem.

Question (c) Why is *E. coli* a poor host for a human protein that requires glycosylation, and what is used instead?

MODEL ANSWER

(a) IPTG (isopropyl beta-D-1-thiogalactopyranoside) is a **non-metabolisable lactose analogue**. It binds and inactivates the **LacI** repressor, but the cell cannot break it down, so induction is **sustained and reproducible** rather than depending on how much lactose happened to be consumed. In minimal or poorly-reducing media, lactose would not induce reliably at all.

(b)

- **Advantage:** no induction step is needed - expression is automatic, which simplifies the process.
- **Problem:** you cannot *tune* expression. Constitutive expression is usually undesirable because it burns the host's resources and may produce toxic intermediate levels.

A better design is an **IPTG-inducible construct**: a *lac operator* (or *lacN*-controlled repressor) placed downstream of the promoter so IPTG gives dose-dependent control - a **tunable** induction rather than an on/off switch.

(c)

- *E. coli* is a **prokaryote** - it has no ER, no Golgi and **no glycosylation machinery**, so it cannot attach the carbohydrate trees that many human proteins need to fold and function.
- Hence **mammalian CHO cells** (Chinese hamster ovary) are used for complex, glycosylated human proteins; **codon optimisation** adapts the gene sequence to the host's codon usage.
- A further reason to delay induction: if expression starts before exponential growth, the burden and misfolding drive **inclusion bodies**.

Common trap: treating the lac operon as three independent inputs. It is a **two-input** gate: **LacI (lactose)** AND **CAP-cAMP (glucose)** both have to permit transcription.

SECTION F - Molecular biology methods (Lecture 6)

Case 17 - The specimen that arrives in the wrong form

Vignette

A pathologist sends a colleague an **FFPE** tumour block for expression profiling. The laboratory extracts RNA and runs the **Bioanalyzer**: the trace is a smear with no clean peaks (**RIN 3** on the slide's own scale, where intact RNA is **RIN 10** and partially degraded RNA is **RIN 5**). The same patient arrives with two blood tubes drawn at one visit - one plain, one containing **EDTA**. A circulating-DNA assay reports *not detected* in the plasma tube but a clearly elevated level in the serum tube from the same person, and an **ELISA** for a plasma protein reads consistently lower in the EDTA tube than in the serum tube.

Question (a) What do you ask the pathologist for, and what would the Bioanalyzer trace of an acceptable sample look like? State the cut-off.

Question (b) The same target reads *undetectable* in plasma and elevated in serum. Explain that, and say what you must state when a question mentions a blood biomarker without naming the matrix.

Question (c) The ELISA reads lower in the EDTA tube. Explain the artefact, and say what it means for how the assay must be validated. Finish with the limit: what can this quality check not tell you?

MODEL ANSWER

(a) Ask for **fresh frozen** tissue, not the FFPE block.

- **Observation:** the Bioanalyzer trace is a smear with no clean peaks, **RIN 3**, which is below the slide's own cut-off of **RIN < 6 = poor quality**.
- **Mechanism:** formalin crosslinks the nucleic acid, so the two ribosomal peaks - **28S** and **18S**, normally in a **2:1** ratio - are degraded away and the RNA does not survive extraction intact.
- **Interpretation:** FFPE is the right material for **histology and IHC**, because the pathologist needs the architecture, and the wrong material for genomic and transcriptomic profiling, because the nucleic acid is degraded. This is the lecturer's own "garbage in, garbage out" frame - no downstream technique recovers information that was destroyed in the sample.
- **Action:** request **fresh frozen** tissue and re-extract. An acceptable RNA sample gives **two sharp peaks at 28S and 18S**. If you are stuck with FFPE, **RIN** is less informative because the ribosomal peaks are lost, so use **DV200** (the >200 nt fraction) as the FFPE metric.
- **Limit:** the quality check tells you the extraction worked; it cannot tell you whether the analyte is *clinically* meaningful.

(b) Serum and plasma are not the same matrix, and the answer can invert between them. Serum is blood collected **without** anticoagulant, left at room temperature to clot, with the supernatant taken; plasma is blood collected **with** anticoagulant (e.g. **EDTA**), centrifuged, supernatant taken. The lecturer's rheumatoid arthritis example makes the point: the same target, the same patients and the same assay gave **no difference at all** in the plasma panel but a **significant elevation in the RA patients** in the serum panel, and he attributes it to the **half-life and stability of the analyte** in the two matrices. The interpretation is that the matrix is part of the biology of the measurement, not a detail of the tube. So the action is to **state the matrix explicitly** - and if a question mentions a blood biomarker without naming the matrix, say that the result cannot be interpreted without it.

(c) The **anticoagulant is poisoning the assay**. EDTA inhibits the enzyme in the ELISA readout, so the signal is reduced - the colour you measure is partly an artefact of the tube you used. Two consequences follow:

- **Interpretation:** the readout is not purely analyte-driven. The matrix is part of the measurement, so the assay must be **validated in the matrix it will actually be run in**, not in a clean buffer.

- **Action:** fix the **tube type** in the protocol so that the anticoagulant is controlled, and make the matrix itself a controlled variable. In a multi-site study, run a **dry run** first: shipping the blood tube lying **horizontal** on a foam board beat shipping it **vertical**, because more gas exchanges with the blood and the cells do not settle and pack against the bottom; the readout is **PBMC viability**.

And the limit: none of this quality control can tell you the analyte is *clinically* meaningful. A trustworthy measurement and a useful measurement are two different things.

Common trap: reporting a blood biomarker without naming the matrix, and blaming the antibody when the tube is the culprit. Serum and plasma can give opposite answers for the same analyte in the same patient.

Case 18 - Which instrument, and which way round the screen reads

Vignette

A university spin-out is running a liver cancer project with three questions: (1) which genes differ between tumour and adjacent liver, including any novel splice isoform; (2) which gene makes the tumour cells resistant to sorafenib; and (3) how much of a candidate soluble protein is in patient sera. Before choosing, they benchmark the two transcriptomic platforms on matched reference samples: the platforms agree well (R ~ 0.44-0.67 across the five treatments of a published comparison), but on the ANIT treatment RNA-Seq returned 3,139 differentially expressed genes against the array's 1,014, with maximum fold-changes of 577.5 against 41 (Rao MS et al., 2019, *Frontiers in Genetics* 9:636). For question (2) they run a pooled **CRISPR/Cas9** knockout screen and sequence the survivors by NGS.

Question (a) For questions (1) and (3), name the technique you would use, state the reason, and state one thing it cannot do.

Question (b) After drug treatment the sgRNAs targeting **gene A** are enriched among the survivors, while the sgRNAs targeting **gene B** are almost absent after drug treatment but abundant in the untreated arm. What does each gene confer, and what is each screen called?

Question (c) Why must the untreated arm be run at all, and what do you do next once you have a hit?

MODEL ANSWER

(a) For (1) use **RNA-Seq**: it sequences everything, so it finds novel transcripts and splice isoforms, and it does not saturate on abundant transcripts. The comparison in the vignette is exactly why the microarray was abandoned - the two platforms agree on *what* is expressed (R ~ 0.44-0.67) but RNA-Seq detects more differentially expressed genes with much larger fold-changes, because the array saturates on abundant transcripts and loses signal on rare ones. Its limit: it does not resolve *which cell type* is expressing - for that you need **scRNA-Seq** - and it is not cheap.

For (3) use an **ELISA**: it is quantitative for a soluble protein and works on serum and plasma. If the antigen is low-abundance, use the **indirect** format, because several enzyme-conjugated secondary antibodies bind each unconjugated primary and the signal is amplified; the direct format gives one antibody and one signal. Its limit: an ELISA measures a *concentration*, not cells - counting antigen-specific cells is **ELISpot** - and it tells you nothing about which isoform or which cell produced the protein. If the candidate turns out to be a **novel** protein, remember that **LC-MS** cannot identify it, because its mass fingerprint must be matched against a database; **LC-MS/MS** fragments the peptide down to two-amino-acid fragments and performs **de novo** sequencing, which is how a novel protein or a novel splice variant is found.

(b) Both screens put the same two things into the cells - the **Cas9** enzyme and the **sgRNA library** - and Cas9's cut is repaired by error-prone **NHEJ**, so the knockout is a genuine loss of function. The difference between the two screens is what happens to the knockout cells after the drug treatment. Lead with the readout, then name the consequence:

- **Gene A - enriched among the survivors.** The knockout cells survive the drug, so **gene A confers drug sensitivity**. This is the **positive screen** (positive selection, FOF).

- **Gene B - depleted after drug treatment, abundant in the untreated arm.** The knockout cells die under the drug, so **gene B confers drug resistance**. This is the **negative screen** (negative selection, dropout).

(c) The untreated arm is what separates a genuine drug-response gene from a gene that is simply essential for survival. Without it you mix up the finding with cell survival itself: if the sgRNAs for **gene C** are lost in the untreated arm as well, gene C is essential for survival and says nothing about the drug.

Next, in this order:

- **Validate the hit** in an **orthogonal assay**: the screen's output is a target hypothesis, and the next step is **target validation** before anything is built on it.
- **Do the enrichment analysis** on the genes that co-vary with the hit (**g:Profiler**, **GSEA**, then **Cytoscape** and **EnrichmentMap** for the network).
- **State the limit:** the screen identifies genes whose loss changes fitness under the drug treatment; it does not tell you the mechanism, and it cannot distinguish a drug-response gene from an essential gene without the control arm.

Common trap: saying that the screen which finds the surviving knockout is the screen that finds the resistance gene. It is the other way round - the knockout that is **enriched** (positive screen) confers **drug sensitivity**, and the knockout that is **depleted** because the cells die (negative screen) confers **drug resistance**. And never quote a hit without the untreated control arm.

SECTION G - Genomic applications in biomedical research (Lecture 7)

Case 19 - A score, a trial, and a treatment decision

Vignette

A woman has **Stage I** invasive breast carcinoma that is **ER-positive / HER2-negative** and **node-negative**, with a tumour **<= 5 cm**. Her tumour is sent for **Oncotype DX** and the **Recurrence Score (RS)** is **24**. On the lecturer's multivariate slide the Recurrence Score is the only factor apart from poorly differentiated grade whose confidence interval excludes 1: **HR 2.81 (95 % CI 1.70-4.64), P < 0.001**, computed **per 50-point increment** of the score; age at surgery (P = 0.22), clinical tumour size (P = 0.38) and HER2 amplification (P = 0.06) were all non-significant. In the randomised **NSABP B-20** trial - 651 node-negative, ER+ patients, 227 on tamoxifen alone and 424 on tamoxifen plus chemotherapy - the benefit of chemotherapy was confined to the high-score group.

Question (a) Is an RS of 24 prognostic, predictive, or both? Justify each answer.

Question (b) What does the randomised trial show in the high, intermediate and low score groups, and what does the mechanism predict? Can chemotherapy be withheld from this patient?

Question (c) Her sister has the same tumour but a **MammaPrint** result of HIGH risk. Can you withhold chemotherapy from her on the strength of that result? What is the difference between a **biomarker** and a **companion diagnostic**, and which of the two assays here is the CDx?

MODEL ANSWER

(a) Both.

- **Prognostic:** the Recurrence Score is an *independent* predictor of distant recurrence in the multivariate analysis - **HR 2.81 (95 % CI 1.70-4.64), P < 0.001**, **per 50-point increment** of the score, while age, tumour size, HER2 amplification and ER level were all non-significant. Note the scale: it is per 50 points, not per point.
- **Predictive:** in the randomised trial with a treatment arm, the score separated the patients who gained from adjuvant chemotherapy from those who did not. A predictive claim needs a treatment arm; a prognostic claim does not.
- **Where 24 falls:** the bands are **< 18 = low**, **18 to < 31 = intermediate**, **31 to 100 = high**, so 24 is *intermediate* risk.

(b) In the randomised trial (Paik et al., J Clin Oncol 2006;24:3726-3734; a retrospective analysis of B-20, interaction test **P = 0.038**):

- **High RS:** chemotherapy significantly improved recurrence-free survival.
- **Low and intermediate RS:** no significant benefit, and in the low-RS patients the **odds ratio for chemotherapy exceeded 1**, which suggests net harm - the toxicity without the anti-tumour benefit.
- **The mechanism:** a high RS means high proliferation and high driver signalling, so the recurrence risk on endocrine therapy alone is high and cytotoxic chemotherapy removes the proliferating cells. A low RS means little intrinsic proliferation, so there is little to kill and the toxicity is all that is left.

So for this patient - **RS 24, intermediate** - the trial shows **no significant benefit** from chemotherapy, and endocrine therapy alone is the defensible choice. The limit: this is the node-negative, ER+ population of B-20, and the benefit was confined to **RS >= 31**; it does not extend to HER2-positive or late-stage disease, which is why the assay is not indicated there.

(c) No. MammaPrint is a **prognostic** signature - **70** genes on a **DNA microarray**, reporting **high risk / low risk** of distant recurrence, and that 70-gene signature was itself validated in an independent cohort of **295** patients (van de Vijver et al., N Engl J Med 2002). On the evidence shown it *does not* tell you who benefits from chemotherapy, and a prognostic assay alone cannot justify withholding a treatment: that is a **predictive** claim, and it can only be demonstrated as a treatment-by-biomarker interaction in a randomised trial. So the sister needs the same predictive evidence, not a different prognostic test.

The two-step concept: the **biomarker** is the biological entity itself - in the lecturer's example **HER2** overexpression on the tumour cell surface, and here the 21-gene expression signature. The **companion diagnostic (CDx)** is the *assay* that measures it in a clinic - in the example **HER2 IHC** or **HER2 qPCR**, and here the **Oncotype DX 21-gene RT-qPCR assay** that actually runs and produces the Recurrence Score. Both are needed, they are not the same thing, and the CDx is developed *alongside* the drug it serves, so it has to be ready before recruitment starts.

Common trap: calling the Recurrence Score prognostic and then using it to withhold chemotherapy, or quoting HR 2.81 as a hazard ratio *per point* rather than **per 50-point increment**. And do not treat MammaPrint and Oncotype DX as two versions of the same test: 70 genes on a microarray with two risk bands, against 21 genes on RT-qPCR with three.

Case 20 - An assay that measures well and means nothing

Vignette

A diagnostics company has taken a candidate plasma protein forward to a clinical study. The laboratory file is complete: the lower limit of quantification is set; the antibody was counter-screened against the homologues **CDH16** and **CDH1**, so only strictly **CDH17**-specific clones were taken forward; the dynamic range was extended to cover the pathological range seen in cancer patients; the signal is reproducible across days, operators and two antibody lots made in **2024** and **2026**; and the matrix effect was tested by spike-in across **three matrices**. In the first patient cohort, however, the assay does not separate cases from controls on the ROC curve, so no cut-off can be chosen, and the programme stops. The same company's breast cancer arm profiles tumour tissue only; the director asks what the tissue **2 cm** from the tumour edge would add.

Question (a) Which validation stage has been passed and which has failed? State each in one sentence, and name the third stage.

Question (b) The biomarker may yet be fixed. If a drug trial has to find the patients who will respond, which trial design would you use, and what has to be ready before recruitment starts? Which design suits a marker expressed in only a small fraction of patients?

Question (c) Use the tumour-adjacent tissue study to say what the breast cancer arm is missing and what it changes.

MODEL ANSWER

(a) The validation ladder has three rungs, and this company climbed the first and fell at the second.

- **Passed - analytical validation:** does the assay measure its target correctly and reproducibly? It needs sensitivity (the **lower limit of quantification, LLoQ**), specificity, dynamic range, linearity, precision, reproducibility (inter-day, lot-to-lot) and matrix effect. That file is complete.
- **Failed - clinical validity:** is the measurement associated with the clinical endpoint? Here it is not, so no ROC cut-off can be chosen.
- **The third rung - clinical utility:** does using the test improve patient management and outcome? Only a clinical trial can show that.

The interpretation is the lecturer's own sentence: analytical precision is only the foundation, and an analytically perfect assay with no clinical utility changes nothing for a patient - a biomarker is useless unless it has clinical utility. The action is to fix the clinical-validity problem or to stop; the limit is that no amount of extra analytical work rescues a marker that does not track the endpoint.

(b) Enrichment design: screen every incoming patient with the validated CDx and enrol only those who test positive. The CDx must be ready *before* recruitment starts, because in this design it is what screens the patients in - it is developed alongside the drug. The reward is a smaller, faster and cheaper trial (the lecturer's own figure is about **US$200,000** per patient enrolled), and the ethical point that it avoids giving a toxic drug to patients who cannot benefit. **All-comer design**: enrol everyone, measure the biomarker retrospectively, and infer its value from the treatment-by-biomarker interaction - this is the design in which a **predictive** biomarker is identified.

The limit to state: the enrichment design carries an initial screening burden and is wasteful when the biomarker is expressed in only a small fraction of patients, which is why nobody uses **beta18** as a biomarker. So the design choice depends on prevalence and on whether a CDx is ready.

(c) The tissue **2 cm** from the tumour edge is histologically normal but molecularly abnormal, and it carries prognostic information of its own: patients whose adjacent tissue carried an **"active"** signature had poorer survival, and those with an **"inactive"** signature did better, with the genes in that favourable signature enriched for **anti-tumour immunity** - an active local immune response that suppresses micrometastases (Roman-Perez et al., Breast Cancer Res 2012;14:R51). In the exam, say what he taught - the favourable "inactive" signature is the immunity-enriched one; know separately that in the paper the Active subtype is the inflammatory, fibrosis-and-macrophage one.

Two things change. In theatre, the operation stops being "cut out this lump" and becomes "treat this field": if the microenvironment is hostile, a narrow margin leaves hostile tissue behind and the recurrence risk persists, so the margin has to be planned with the biology of the adjacent tissue in mind - and a patient with a favourable adjacent signature is a lower-risk patient even though the tumour looks identical. In the laboratory, the instruction is the one the lecturer chose to end the lecture on: do not only look at the cancer tissue, but also at the adjacent tissue in your future bioinformatics analyses. The limit: the adjacent-tissue signature is *prognostic*, not *predictive* - it tells you who is at risk, not which drug to give.

Common trap: treating a clean analytical file as proof that the test works, and calling a prognostic signature a basis for withholding a treatment. Only clinical validity and clinical utility change anything for a patient.

Self-marking scale

Use this when a case is close:

Marks	What earns them
1 mark	Correct drug / technique named.
2 marks	Correct mechanism stated (which pathway, which protein, why).

Marks	What earns them
3 marks	Correct mechanism **and** the clinical consequence - the "so what for the patient".
4 marks	All of the above **plus** naming the biomarker type (predictive / prognostic / surrogate).

The lecturer's own words: *"multiple choice is for you to pass the exam, but if you want to get a full mark or a better result, you'd better do perform well for the short questions, which emphasize on data analysis and also the scenario questions."*

Every case above is a **scenario question with data**. Answer in **English**.

GNBF5020 Revision Cheat Sheet

Cross-lecture master summary · English edition

Cross-lecture master summary - Dr. Tony Wong - 7 lectures (12 Sep - 7 Nov 2026)

A condensed, last-minute review of the highest-yield concepts and clinical linkages. For full detail (audio narrative + slide content merged) see:

Per-lecture complete packs: `Lecture_1_Complete_Revision.md` ... `Lecture_7_Complete_Revision.md`

Single master document (all 7 lectures, audio + slides): `GNBF5020_Complete_Revision_All_Lectures.md`

Audio-only notes (legacy, Lectures 1-5): `Lecture_1_Revision_Notes.md` ... `Lecture_5_Revision_Notes.md`

Slide-only supplements (legacy, Lectures 1-5): `Lecture_1_Slides_Supplement.md` ... `Lecture_5_Slides_Supplement.md`

Lectures 6-7 exam-focus guides: `GNBF5020_Lecture_6_Exam_Focus_and_Learning_Guide.md` - `GNBF5020_Lecture_7_Exam_Focus_and_Learning_Guide.md`

Case study bank: `GNBF5020_Case_Studies.md` **- 20 exam-style scenarios with model answers**

[target] Two course-wide themes (every lecture touches them)

- **Diversity** - genetic variation, antibody variability, alternative splicing, meiotic recombination, tumour heterogeneity, single-cell heterogeneity.
- **Stability** - DNA repair, mitochondrial maternal inheritance, epigenetic cellular memory, regulatory checkpoints in drug development, sample and assay quality control.

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1. Lecture 1 - Introduction to Biomolecules

Four biomolecule classes + clinical/biotech applications

Class	Key concept	Clinical / exam-relevant example
Carbohydrates	mono-/di-/polysaccharides	Glucose-6-phosphatase deficiency -> glycogen storage; **Warburg effect** (aerobic glycolysis in cancer)
Lipids	saturated vs unsaturated FAs -> triglycerides / phospholipids / bilayer	**LPS -> TLR4 -> NF-kappaB** -> septic shock
Proteins	Primary (sequence) -> Secondary (alpha-helix/beta-sheet) -> Tertiary -> Quaternary (subunits); **domain** = independent fold unit	Edman degradation (legacy) / LC-MS/MS (modern); **deamidation** (Asn->Asp) liability in biologics
Nucleic acids	Nucleoside = base + sugar; **nucleotide** + phosphate	**Avery-MacLeod-McCarty** proved DNA = transforming principle; Watson/Crick/Franklin = double helix

Cancer case studies to remember:

- **Warburg effect** = cancer lactate even with O2; intermediate diversion -> biosynthesis.
- **Hippo pathway** = MST1 -> LATS1 -> YAP phosphorylation -> YAP excluded from nucleus -> no proliferation. *MST1* KO -> YAP nucleus -> overgrowth.

- **Kaplan-Meier lesson**: same target can be **pro-tumour in one cancer, anti-tumour in another** (e.g., RTK high = poor in pancreatic, *better* in renal cell).
- **K-Ras**: constitutively active when mutated -> **cetuximab fails** in mutant K-Ras CRC. K-Ras was "undruggable" -> now FDA-approved small-molecule (pancreatic).
- **miRNAs** are exceptionally **stable in serum** -> preferred non-invasive biomarkers.

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2. Lecture 2 - Basic Cell Biology

Tools, organelles, division, stem cells, immunotherapy

Analytical toolbox:

- **Flow cytometry** -> FSC = size, SSC = granularity; **gating** isolates subpopulations.
- **FACS** -> physically sorts viable cells (sterile hood) for downstream work.
- **LCM** -> pure tissue region via laser; avoids stromal/immune contamination.
- **FISH** -> DNA probes, not antibodies -> **HER2 amplification** test.
- **IF + confocal** -> multiplex, 3D, subcellular resolution.

Mitochondria - three things to remember:

- **Endosymbiotic origin** (invaded bacteria).
- **Maternal inheritance** only (paternal mtDNA destroyed post-fertilisation) -> "Mitochondrial Eve".
- **Threshold effect** in disease: ratio of mutant mtDNA must exceed tissue-specific tolerance.

Cell cycle: G1 -> S -> G2 -> M -> cytokinesis; **cyclin-CDK** drives transitions; **PI staining + flow cytometry** quantifies phase; **serum starvation** synchronises.

iPSC reprogramming (Yamanaka factors Oct3/4, Sox2, Klf4, c-Myc):

- Validation gold-standard: **teratoma assay** in mice -> all 3 germ layers.
- Uses: disease modelling, drug screening (e.g., cardiotoxicity on iPSC-cardiomyocytes), regenerative medicine.

Therapeutic antibodies & engineering:

- **IgG1 -> ADCC** (Fc engages NK cells); **IgG4 -> no ADCC, longer half-life**.
- **CAR-T** = scFv + intracellular T-cell activating domains (CD3zeta, 4-1BB, CD28).
- **Autologous** = patient-derived; **allogeneic** = donor/iPSC; **in vivo** = mRNA-LNP-delivered.
- **CTCs (EpCAM+ / CD45-)** = liquid biopsy, but watch out for **EMT** changing markers.

MSI-H tumour biology (sets up Lecture 4): defective MMR -> frameshift neoantigens -> "hot tumour" -> responsive to **anti-PD-1/PD-L1**.

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3. Lecture 3 - DNA and Chromatin Structure

Packaging -> chromatin state -> epigenetic disease

Hierarchy: DNA wraps histone octamers -> **nucleosomes** -> chromatin fibre -> chromosome.

- **Euchromatin** = open, transcribed. **Heterochromatin** = silent, condensed.
- **ATP-dependent remodelling complexes** + nucleosome sliding determine accessibility.

Cytogenetics:

- **G-banding** detects >= 0.5-10 Mb changes - misses point mutations.
- Centromere = spindle attachment; p-arm (short), q-arm (long).
- **Aneuploidy**: non-disjunction in meiosis -> Trisomy 21 (Down), Turner 45,X.
- **Inversion** doesn't change gene dosage but **moves genes away from their enhancers**.

Epigenetics - the high-yield layer:

- **CpG methylation** -> silenced promoters (esp. tumour suppressors).
- **Histone code nomenclature** = `H3K9me3` (Histone 3, Lysine-9, tri-methylation).
- **DNA methylation <-> histone marks** = cross-talk -> cellular memory.
- **Bisulfite sequencing** = gold-standard methylation profiling (bisulfite: C -> U unless methylated).

Clinical methylation biomarkers (KEEP THESE):

- **MGMT promoter methylation** -> silences DNA repair in glioblastoma -> **sensitises to alkylating chemo + RT** -> better survival.
- **CIMP (CpG Island Methylator Phenotype)** in colorectal cancer -> worse prognosis.

Telomeres:

- Shorten each division -> **Hayflick limit** -> senescence.
- ESCs / select adult stem cells / cancer cells maintain via **telomerase** (reverse transcriptase with internal RNA template).
- In HCC: chronic liver injury + MYC-driven telomerase reactivation -> re-lengthening -> **prognostic marker**.

CNVs: variable copy numbers in the population; *AMY1* amylase = textbook dietary CNV; mechanism = **NAHR** (non-allelic homologous recombination). Distinguish **driver** (causal) from **passenger** (co-amplified) genes by integrating CNV + mRNA expression profiles (case: *NPM1* in leukaemia).

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4. Lecture 4 - DNA Replication, Repair and Recombination

One cycle: replication -> damage -> repair -> recombination -> therapy

Replication enzymology:

- **Helicase** (unwind) -> **topoisomerase** (relieve torque) -> **primase** (RNA primer) -> **DNA polymerase** (5'->3' + 3'->5' proofreading) -> **ligase** (seal nicks).
- **Anti-parallel** -> continuous **leading strand**, discontinuous **lagging strand** as **Okazaki fragments**.
- **Eukaryotes** use **thousands of replication origins** forming bidirectional **replication bubbles**.

Damage classification (small -> large):

Pathway	What it fixes	Key enzyme / concept
BER	single damaged base	glycosylase -> AP endonuclease -> polymerase -> ligase
NER	bulky multi-nucleotide lesions	e.g., UV pyrimidine dimers
MMR	replication mismatches + small indels	**microsatellites** <-> **MSI-H**

Pathway	What it fixes	Key enzyme / concept
HR	DSBs (high-fidelity; needs sister chromatid)	S/G2 only
NHEJ	DSBs (error-prone; no template)	works all phases; **basis of V(D)J recombination**
DDR sensors	ATM, ATR	trigger repair or apoptosis via **p53**

Pharmacology hooks (high-yield):

- **PARP inhibitors** = synthetic lethality in **BRCA1/2-deficient** tumours (HR deficient) -> wide therapeutic window.
- **Topoisomerase-inhibitor ADCs** (trastuzumab-deruxtecan, TROP2 ADCs).
- **Dual-payload ADCs** (topo + DDR inhibitor) defeat single-drug resistance.
- **MSI-H** -> anti-PD-1/PD-L1; **CEA** = surrogate circulating biomarker in CRC.
- **Proofreading loss** -> high neoantigen burden -> immunotherapy-receptive.
- **CRISPR/Cas9** -> DSB -> repaired by NHEJ (knockout) or HR (knock-in). HR works only in S/G2.

IP/FTO: Berkeley (Doudna/Charpentier) vs Broad (Zhang) CRISPR patent dispute illustrates **Freedom-to-Operate** analysis required before commercialising.

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5. Lecture 5 - Control of Gene Expression

Every layer is a drug target or biomarker

Prokaryotic paradigms:

- **Lac operon** (inducible): lactose -> allolactose -> LacI repressor off; low glucose -> ^cAMP -> CAP binds -> strong transcription. IPTG = non-metabolised inducer for recombinant *E. coli* systems.
- **Trp operon** (repressible): tryptophan binds Trp repressor -> shut down biosynthesis.

Eukaryotic transcriptional control:

- **Pol II + general TFs** at **TATA box**; **enhancers/silencers** can sit thousands of bp away -> DNA-bending proteins loop them onto the promoter.
- **Reporter assay** (luciferase deletion series) = standard way to map promoter elements.

Nuclear receptors (e.g., estrogen receptor, androgen receptor):

- Architecture: **NTD / DBD / LBD**.
- **Dimerisation** dramatically increases DNA-binding affinity (HER2 heterodimerises; Herceptin blocks this).
- **Tamoxifen** = SERM: antagonist in breast, partial agonist in endometrium -> tissue-selective pharmacology.
- **Receptor occupancy** required for clinical effect differs by cancer type (HER2: 2% breast vs 10% gastric).

Post-transcriptional regulation:

- **5' cap**, **poly-A tail length**, **3' UTR** -> mRNA stability & translation.
- **Alternative splicing** -> multiple isoforms from one gene; in cancer: produces **neoantigens** and **soluble biomarkers**; single-nt mutations can create new splice sites (HCC case: short splice variant -> poorer survival).
- **miRNAs (~22 nt)** bind 3' UTR seed sequences -> mRNA degradation or translational repression; **stable in serum** -> non-invasive biomarkers.

Signalling pathways -> nuclear TFs:

- **NF-kappaB**: held by **IkappaB**; LPS -> IkappaB degraded -> NF-kappaB nucleus -> inflammation. Cytokine temporal order: **TNF-alpha -> IL-1beta -> IFN-gamma -> IL-2 -> IL-15**. Positive-feedback loop -> rapid defence.
- **HIF-1alpha**: degraded normoxia; stabilised hypoxia -> binds **HREs** -> induces **VEGF**. **Synthetic switch:** HRE-controlled CAR expression -> activates only in hypoxic tumour.
- **Hippo-YAP**: MST1/LATS1 -> YAP phosphorylation -> excluded. Loss-of-function -> YAP nucleus -> overgrowth -> cancer.

Modern applications:

- **CAR-NK / CAR-M** - better solid-tumour infiltration than CAR-T.
- **synNotch AND-gate CARs** - require two antigens -> high specificity.
- **mRNA vaccines** - exogenous mRNA -> translated -> MHC-presented -> activates T/B-cells. Personalised by single-cell RNA/TCR sequencing of patient neoantigens.
- **lncRNAs** - non-coding, scaffold/guide for chromatin machinery.

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6. Lecture 6 - Methods in Molecular Biology

Sample first -> nucleic acids -> many genes at once -> proteins -> perturb and express

Block	Key concept	Clinical / exam-relevant example
The sample gates everything	FFPE vs fresh frozen; serum vs plasma; homogeneity; central lab + dry run	"Garbage in, garbage out"; RA biomarker - no difference in plasma, significant elevation in serum; **EDTA** inhibits the ELISA reporter enzyme -> lower signal
RNA quality gate	Bioanalyzer **28S : 18S ~ 2 : 1**	Two sharp peaks = intact RNA; FFPE gives a smear; **RIN < 6 = poor** (RIN 10 = intact)
Real-time PCR	RNA -> cDNA -> PCR; **Ct** = threshold cycle	**Ct ~13-17** for house-keeping genes; **larger Ct = LOWER copy number**; relative expression by DCt
SYBR Green vs TaqMan	SYBR Green binds any dsDNA (cheap, not sequence-specific); TaqMan = primer + **probe** (reporter **Q**)	TaqMan is specific; multiplex = several targets in one reaction -> less between-test variation, less sample
Microarray vs RNA-Seq	known sequences by hybridisation vs sequence everything	R ~ 0.44-0.67 (Rao et al., *Frontiers in Genetics* 2019); ANIT 1014 vs 3139 DEGs; MDA max fold-change 15.7 vs 226.5
Bulk vs scRNA-Seq	bulk averages every cell; scRNA-Seq = one cell, one transcriptome	21 cancer types, 316 patients, **397,810 T cells** (Zheng L et al., 2021)
ChIP-Seq	antibody pulls down the DNA-protein complex -> sequence it	maps where TEAD binds - a hypothesis until validated with a **luciferase reporter assay**

Block	Key concept	Clinical / exam-relevant example
Protein separation	SDS-PAGE (size) - IEF (pI) - 2D-PAGE	IgG reducing -> ~50 kDa heavy + ~25 kDa light; non-reducing -> ~150 kDa intact
Mass spectrometry	LC-MS = fingerprint vs a database; LC-MS/MS = de novo	LC-MS **cannot** identify a novel protein; MS/MS fragments down to 2 amino acids
Immunoassays	direct vs indirect ELISA; ELISA vs ELISpot	indirect = **signal amplification** for low-abundance antigen; ELISpot counts IFN-gamma-secreting cells; multiplex = 10-20 proteins
CRISPR screen	positive vs negative	positive = knockout **enriched** among survivors -> drug **sensitivity**; negative = **depleted** -> drug **resistance**
Expression in cells	transfection vs electroporation; Fc vs GFP	lipid transfection fails on T/NK/macrophages -> **electroporation**; Fc forces **dimerisation** (nuclear receptors, HER2); Gly3Ser linker

Exam points to nail:

- **Positive vs negative CRISPR screen** (the question he named): both give the cells **Cas9** + an **sgRNA library**; **positive** = knockout **enriched** among survivors -> the gene confers drug **sensitivity**; **negative** = knockout **depleted** because the cells die -> the gene confers drug **resistance**. The untreated control arm separates genes that are simply essential for survival from genes that modify the drug response.
- **NHEJ, not MMR**: defective **NHEJ** -> no **V(D)J recombination** -> no **TCR repertoire** *and* no antibody repertoire (B cell / plasma cell) -> **SCID**. Name both limbs.
- **Describe vs explain**: "describe" earns the data, *explain* earns the mechanism - naming the mechanism is the part students omit. Answer order: observation -> mechanism -> clinical meaning -> limit.
- **The sample is the first variable**: FFPE is for histology/IHC only; serum vs plasma can invert a result; one central lab removes operator/day/site variation; a dry run ships fake samples under different conditions (readout = PBMC viability).
- **Ct direction**: larger Ct = **lower** starting copy number; small Ct = abundant target.
- **Know what each technique cannot do**: ChIP-Seq maps binding but must be validated with a reporter assay; microarray cannot find a novel transcript or isoform; LC-MS cannot identify a novel protein; bulk RNA-Seq cannot tell you which cell type is expressing.
- **Do not memorise**: platform mechanics, LC instrumentation, how NGS machines are built - but do know *what each technique measures and what it therefore cannot tell you*. Microarray is not dead: **MammaPrint** (70 genes, FDA-cleared) is still in clinical use.

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7. Lecture 7 - Genomic Applications in Biomedical Research

Omics data -> pathway -> target -> drug -> test -> score -> treatment decision

Pipeline step	Key concept	Clinical / exam-relevant example
Omics -> pathway	**PEA**: g:Profiler (over-representation), GSEA (ranked list), Cytoscape + EnrichmentMap	a DEG list is not an answer; genes act in pathways, not alone
Pathway -> target	driver vs passenger	driver = **up-regulation + copy-number change + enriched pathway**; expression alone is confounded
Drug -> test	biomarker vs **CDx**	biomarker = HER2 over-expression; CDx = **HER2 IHC** or **HER2 qPCR**, developed **alongside** the drug
Finding the patient	qPCR vs ddPCR	qPCR needs a standard curve (10, 5, 2.5, 1.25 ng); ddPCR = thousands of droplets + **Poisson**, no standard curve -> low-abundance **ctDNA** (e.g. **EGFR T790M**)
Test -> score	**MammaPrint** (70 genes, microarray)	high/low risk of distant recurrence; **prognostic only**; validated in an independent cohort, n = 295 (van de Vijver et al., *N Engl J Med* 2002)
Score -> decision	**Oncotype DX** (21 genes = 16 + 5 reference, RT-qPCR) -> **RS 0-100**	bands **< 18 / 18 to < 31 / 31 to 100**; **HR 2.81 (1.70-4.64) per 50-point increment, P < 0.001**
Who benefits	NSABP B-20	**651** patients (227 tamoxifen vs 424 tamoxifen + chemo); benefit only in **high RS**; low-RS odds ratio for chemo **> 1** (Paik et al., *J Clin Oncol* 2006;24:3726-3734)
Beyond the tumour	tissue **2 cm** from the tumour edge	"active" signature = worse survival; "inactive" = better, enriched for **anti-tumour immunity** (Roman-Perez et al., *Breast Cancer Res* 2012;14:R51)
A biomarker in blood	circulating miRNA	**miR-15b + miR-130b**: AUC 0.981, sensitivity 98.25 %, specificity 91.53 %; n = 96 discovery, 15 patients sampled before and after surgery
Validation ladder	**BEST** (Biomarker, EndpointS, and other Tools - an FDA-NIH resource); analytical -> clinical validity -> clinical utility	sensitivity / **LLoQ**, specificity, dynamic range, reproducibility, matrix effect; ROC curve; utility = does using it improve outcome
Trial design	enrichment design vs all-comer design	enrichment screens by biomarker **before** enrolment (needs a ready CDx); all-comer measures it retrospectively; **US$200,000** per patient enrolled, **US$1 million** per IND

Pipeline step	Key concept	Clinical / exam-relevant example
Pharmacogenomics	FDA drug labelling	**CYP2C19** -> abrocitinib (**NDA 213871**): poor metabolisers start at **50 mg once daily**, escalating to 100 mg after 12 weeks

Exam points to nail:

- **Biomarker != CDx**: the biomarker is the biology (a gene, a protein, a signature); the CDx is the assay that measures it in a clinic. A biomarker without a CDx is an observation nobody can act on.
- **Prognostic vs predictive**: prognostic = outcome whatever treatment you give; predictive = benefit from *this* drug, and it needs a **treatment arm** (a treatment-by-biomarker interaction). MammaPrint is prognostic only, Oncotype DX is both - a prognostic assay alone cannot justify withholding a treatment.
- **Validate in an independent cohort**: a signature found in a retrospective research cohort is a hypothesis; it needs a separately recruited independent cohort (MammaPrint: n = 295).
- **Adjacent tissue**: profile the histologically normal tissue **2 cm** from the tumour edge - it carries its own prognostic signature and changes the surgical margin decision.
- **Quote these numbers exactly**: **HR 2.81 is per 50-point increment of RS**, not per point; the chemo-benefit trial is **Paik et al., J Clin Oncol 2006;24:3726-3734**; the invasion genes are **MMP11** (protein alias **ST3**) and **CTSL2**, not CATL2; **BEST** is the FDA-NIH resource, not NCI.
- **The 21-gene panel**: do not memorise the gene IDs - know the five groups (proliferation, estrogen signalling, HER2 signalling, invasion, other).
- **Development is a team sport**: discovery scientists, business development, medical writers, QA/GLP, clinical operations, regulatory affairs, project manager - and a **pre-IND meeting** before the IND submission.

[warning] Cross-cutting key terms to never confuse on the exam

Pair	Difference
BER vs NER	BER = single base; NER = bulky lesions
MMR vs HR	MMR = mismatches/indels; HR = high-fidelity DSB repair
HR vs NHEJ	HR uses template, S/G2 only; NHEJ no template, all phases
cis vs trans	cis = same DNA molecule; trans = diffusible TF from elsewhere
IgG1 vs IgG4	IgG1 = ADCC; IgG4 = no ADCC, longer half-life
Autologous vs Allogeneic CAR-T	patient-derived vs donor-derived
Driver vs Passenger mutation	functional cause vs bystander
Eu- vs Heterochromatin	active vs silent
DBD vs LBD	DNA-binding vs ligand-binding domain of nuclear receptor
de novo vs maintenance methylation	new mark vs copy on daughter strand
ADCC vs CDC	antibody-dependent cellular cytotoxicity vs complement-dependent cytotoxicity
Predictive vs Prognostic biomarker	predicts *response to therapy* vs predicts *disease outcome*
Bulk RNA-Seq vs scRNA-Seq	bulk = average of every cell; scRNA-Seq = one cell, one transcriptome
Positive vs negative CRISPR screen	positive = knockout enriched among survivors -> drug sensitivity; negative = depleted -> drug resistance
Biomarker vs companion diagnostic	biomarker = the biology; CDx = the assay that measures it

[books] Most-tested scenario-question archetypes (per lecturer)

- **Genetic counsellor interpretation** - given a pedigree + BRCA test, calculate proband risk and explain why mutation != certainty (gene-expression control).
- **Kaplan-Meier curve reading** - same target, opposite prognosis depending on cancer type.
- **Drug-target match** - given a patient's K-Ras status, recommend or exclude cetuximab.
- **ASO / siRNA / CRISPR** - pick the right tool for transient knockdown vs stable knockout vs precise knock-in.
- **MSI-H vs MSS CRC** - recommend immunotherapy or not; explain CEA's role as a surrogate marker.
- **Drug liability** - Asn-deamidation in a monoclonal antibody biologic; what to monitor.
- **Clinical trial design** - Phase 1A vs 1B vs 2 vs 3; surrogate vs clinical endpoint; ~$200k/patient cost.
- **CAR-T vs CAR-NK vs CAR-M** - match to solid-tumour vs haematological indication.
- **Tissue-specific pharmacology** - SERM antagonist here, agonist there (tamoxifen).
- **Receptor occupancy** - same HER2 drug, different RO thresholds in breast vs gastric cancer.

- **Positive vs negative CRISPR screen** - given sgRNAs enriched or depleted after drug treatment, name the gene's role and say what the untreated control arm is for.
- **Biomarker vs companion diagnostic** - given a drug and a biomarker, name the assay that finds the patient and say whether the claim is prognostic or predictive.

[calendar] Lecture metadata at a glance

No.	Date	Title	Duration
1	12 Sep 2026	Introduction to Biomolecules	~2h 24m
2	19 Sep 2026	Basic Cell Biology	~2h 20m
3	3 Oct 2026	DNA and Chromatin Structure	~2h 6m
4	10 Oct 2026	DNA Replication, Repair and Recombination	~2h 18m
5	17 Oct 2026	Control of Gene Expression	~2h 11m
6	31 Oct 2026	Methods in Molecular Biology	~2h 18m
7	7 Nov 2026	Genomic Applications in Biomedical Research	~2h 27m

Good luck on the midterm - answer in **English**, defend your reasoning, not your facts. [target]